



Tindeche Alexandru



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Education

2024-2026 Master of Science in Computer Science, Specialization in Artificial Intelligence, Faculty of Engineering, University of Freiburg, Germany

2025: Compulsory course in the Faculty of Economics with Prof. Dr. Dirk Neumann, University of Freiburg: Developed foundations in strategic decision-making and negotiation, game theory, competition and incentives, value and market design, auctions, platform economics, and the functioning of financial, energy, and digital markets.

2021-2024: Bachelor of Science in Computer Science, Mathematics & Computer Science Faculty, University of Bucharest, Romania – Final grade: 9.37 (out of 10)

2017-2021: Student, “Mihai Viteazul” National College, Mathematics and Computer Science (Intensive Computer Science), Bucharest, Romania - Final grade: 9.91 (out of 10)

Work Experience

2025–Present: Master Study Project & Master Thesis — University of Freiburg

- Developed an embedded Edge AI system for real-time detection of ultrasonic plant stress events.
- Integrated Raspberry Pi 5, ADCs, MEMS microphones, and Python-based signal processing for acquisition, event detection, and storage.
- Designed and evaluated supervised, unsupervised, clustering, and semi-supervised anomaly detection (Deep SAD) methods to separate plant events from background noise.
- Built hands-on expertise in Edge AI, ML, signal processing, embedded Linux, hardware/software integration, and experimental validation.

2025: Research Assistant, Cluster of Excellence livMatS, University of Freiburg, FIT Institute, Neptun Labs

- Developed an intelligent hydronic floor-heating prototype for optimizing energy use and enhancing comfort through the predictive power of AI within a versatile control framework

2023: founder member of the Renaissance NGO

- Together with a group of ambitious friends, we are setting the ground to build an open educational platform where people can learn essential topics that are not taught in school.

2023: R&D Internship @ [Keysight Romania](#)

- Developed an integrated solution to automate vulnerability security scans on different network equipment produced by Keysight. Worked in an AGILE environment with weekly SCRUM meetings and coordinated with internal and external customers
- Technologies: Advanced Python OOP, Robot Framework, Selenium, REST API, Office365 API, Networking

2020-2023: Teaching Assistant @ [#CodeSinaia](#) Advanced Programming Summer School,

- Teaching assistant, 2020 Summer School website development, Program Ambassador in social media

Projects

2024: Final BSc Thesis: Automatic traffic analysis using computer vision

- My bachelor's thesis focuses on leveraging state-of-the-art Artificial Intelligence to analyse traffic dynamics at urban intersections. The developed SW application ingests recorded video streams from intersection-mounted cameras, employing advanced CV algorithms (object detection, re-identification, tracking) and complex data visualization and clustering techniques to generate traffic models and detect anomalies.

2023: "Double-Double Dominoes" game score calculator from images of the board using Computer Vision Techniques

2023: Start-viitor , C#, ASP.NET Core MVC, Razor, Bootstrap

- **"Romania fara dosar" Civis Student Hackathon project**, aimed at **addressing societal challenges**. Web portal team work that acts as a digital gateway to public institutions.

2023: Nitro Language Processing

- Python ML application to **detect offensive language in texts posted on different social media** applications

2022: Full Stack Forum App, ASP.NET Core MVC, Razor

Contests

2023: SmartHack Student Hackaton@ University of Bucharest, sponsored by Veridion, big data application dedicated to help investors diversifying their portfolio using a similarity score formula

2023: Deep Image Classification Competition @ University of Bucharest, aim to develop an ML solution that correctly assigns one of the 90 labels to AI generated images, using VGG-16 type of convolutional network

2023: Start-viitor, Student Hackathon "Romania fara dosar", civis project organized by Aspen Institute Romania

2023: Nitro Language Processing Hackaton, SaRoCo: Detecting offensive comments in a Novel Romanian Corpus of News Articles, Python for Machine Learning

2020: American Computer Science League (ACSL) finalist

Skills

Digital skills: Advanced user of MS office suite, Adobe Lightroom, Adobe Photoshop

Programming & Markup Languages: C/C++, C#, Python, Java, JavaScript, HTML, CSS, SQL, ASP.NET Core + Razor

Tools: Git, LaTeX

Familiar with: Object-Oriented Programming Concepts, Edge AI, Signal Processing, Computer Vision, Linux

Training courses

Accreditation training course obtained on UDEMY, CodeCademy, Khan Academy

- Different programming courses, like "Android Java Masterclass", "Complete C# Unity Developer 2D and 3D: Learn to Code Making Games" on Udemy, or "College Computer Science Principles" on Khan Academy, React and JavaScript

Certificates

Udemy: Modern React with Redux

ECDL Computing – ECDL Romania: Programming in Python

Extracurricular

2023: Informatics Tutoring, helping students understand and apply college-level informatics

2019: Bucharest Model NATO & MUNOB – Model of United Nations, debate and diplomacy for high school students

Languages

English – fluent (IELTS C1), French – A2, Romanian – native, German - A2

Other skills

Communication skills: Good communication skills and ability to work with people from different cultural backgrounds, gained throughout my experience as a member-state delegate at the MUNOB and Bucharest Model NATO sessions and Public Speaking training.

Organisational skills: Ability to plan effectively, to coordinate a team, to identify problems and propose resolutions